Last (Family) name: _____

First (Given) name: _____

Student ID (10 digits): _____

University of Toronto
Faculty of Arts & Science

DECEMBER 2024 EXAMINATIONS

STA314H1F
Statistical Methods for Machine Learning I

Duration: 2 hours
Aids Allowed: None

Exam Reminders:

- Fill out your name and student number on the top of this page.
- Ensure you fill out the scantron in the end of the exam booklet.
- Do not begin writing the actual exam until the announcements have ended and the Exam Facilitator has started the exam.
- No aid is permitted. If you possess an unauthorized aid during an exam, you may be charged with an academic offence.
- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials away from your desk. If it is left in your pocket, it may be an academic offence.
- When you are done your exam, raise your hand for someone to come and collect your exam. Do not collect your bag and jacket before your exam is handed in.
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.
- In the event of a fire alarm, do not check your cell phone when escorted outside.

Special Instructions: There are 4 problems in total. Answer Problems 1 - 3 on the exam paper. Answer Problem 4, the multiple choices questions, on the scantron sheet.

Exam Format and Grading Scheme: The exam has 40 points in total. Each sub-question has its points indicated.

Students must hand in all examination materials at the end

Problem 1 (13 points)

Consider the problem of classifying a random, binary, class label $Y \in \{-1, 1\}$ based on a 2-dimensional feature $X = (X_1, X_2) \in \mathbb{R}^2$. We assume $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = 0.5$ and

$$X \mid Y = -1 \sim N(\mu_-, I_2), \quad X \mid Y = 1 \sim N(\mu_+, I_2). \tag{1}$$

Here $\mu_+, \mu_- \in \mathbb{R}^2$ are the mean parameters while I_2 is the 2×2 identity matrix. Answer the following questions.

Recall that the probability density function of a d-dimensional Gaussian $N(\mu, I_d)$ at $x \in \mathbb{R}^d$ is

$$f(x) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\|x - \mu\|^2\right)$$

where $\|\cdot\|$ is the ℓ_2-norm such that

$$\|v\|^2 = \sum_{j=1}^{d} v_j^2 = v^\top v$$

for any vector $v \in \mathbb{R}^d$.

(a) (2 points) Write down the posterior probabilities of $\mathbb{P}(Y = 1 \mid X = x)$ in terms of μ_+ and μ_-.

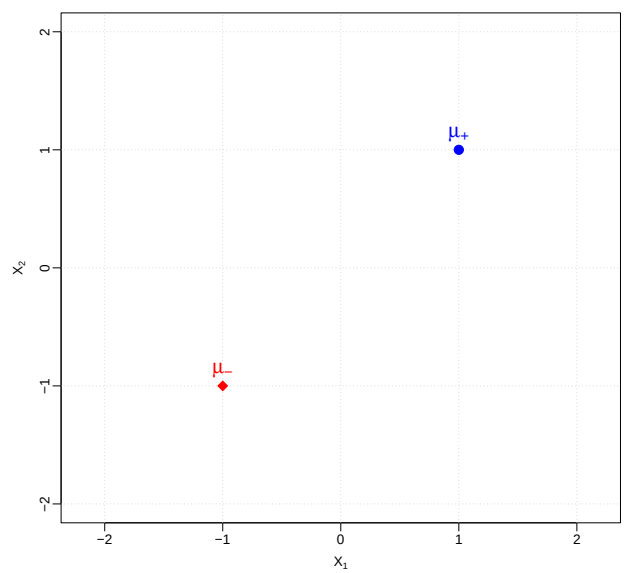
(b) (2 points) Prove that the Bayes classifier f^* is

$$f^*(x) = \begin{cases} 1 & \text{if } \|x - \mu_+\|^2 \leq \|x - \mu_-\|^2 \\ -1 & \text{otherwise} \end{cases}, \quad \text{for all } x \in \mathbb{R}.$$

Briefly explain your geometric interpretation of this classifer.

(c) (1 point) Derive the mathematical expression of the decision boundary of the Bayes classifier.

Suppose $\mu_+ = (1, 1)$ and $\mu_- = (-1, -1)$ indicated in the figure below. Draw your derived decision boundary in the figure (you should also indicate the label classified on each side.)



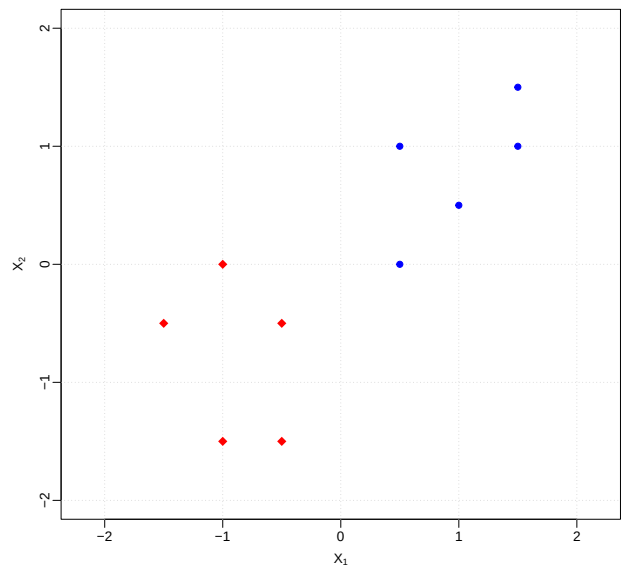
(d) (2 points) Can the binary logistic regression (parametrized by $\beta_0 \in \mathbb{R}$ and $\beta_1 \in \mathbb{R}^2$) be used to estimate the decision boundary in the previous part? If so, state your reasoning and indicate what β_0 and β_1 correspond to. Otherwise, explain why.

(e) (2 points) Suppose the training data contains 10 points:

	$Y = 1$					$Y = -1$				
X_1	0.5	0.5	1.5	1.5	1	-0.5	-1.0	-1.0	-2	-0.5
X_2	1.0	0.0	1.0	1.5	0.5	-0.5	-0.0	-1.5	-0.5	-1.5

Compute the MLE estimates of μ_+ and μ_-. (You may directly use the MLE formula without proving it.)

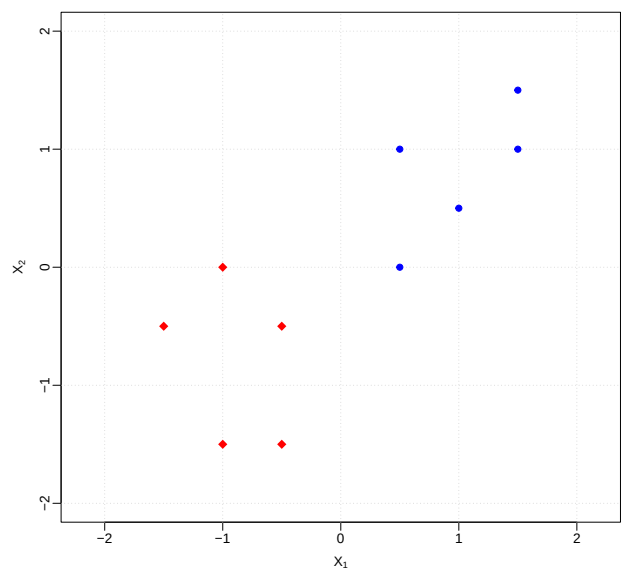
(f) (1 point) Suppose we estimate the Bayes classifier by $\widehat{f}$, the plug-in estimator based on the MLEs of μ_+ and μ_- . Draw its decision boundary on the figure below (the dots correspond to the 10 data points given above).



(g) (3 points) Based on the given training data in part (e), state the decision boundary obtained using the support vector machine (SVM)?

Draw it on the figure and circle the support vectors.

Also compare it with the estimated decision boundary in part (f): which one estimates the Bayes decision boundary better?



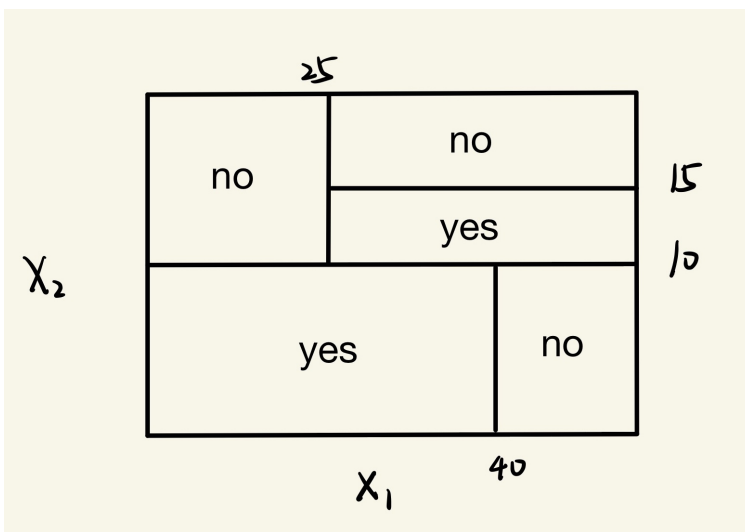
Problem 2 (9 points)

Consider the task of predicting whether a customer will purchase a product based on the following features of the customer:

- X_1 : Age (numerical)
- X_2 : Income (numerical)
- X_3 : Marital Status (categorical: Single, Married, Divorced)

The response is Y = yes if a purchase is made, and Y = no otherwise.

(a) Suppose the fitted decision trees of only using X_1 and X_2 looks as follows.



(a1) (2 points) Sketch the fitted decision tree corresponding to the partition of the feature space.

(a2) (2 points) How many splits are in the fitted decision tree? Is there sufficient information to find a unique order of these splits? If so, give the order. Otherwise, state all possible orders of splits.

(b) Now let us include X_3 (Marital Status) as one of the features. Suppose X_3 is the first feature to split in fitting the decision tree. To figure out how to split, we need to be able to evaluate any given partition (R_1, R_2) of the feature space. Two possible metrics are Gini index and misclassification error rate.

Recall that for a multi-class situation with $Y \in C = \{0, 1, 2, \ldots, K\}$, the Gini index of a given region R is defined as

$$G(R) = \sum_{k \in C} \widehat{p}_k (1 - \widehat{p}_k)$$

where, for any $k \in C$,

$$\widehat{p}_k = \frac{1}{|R|} \sum_{i: x_i \in R} 1\{y_i = k\}$$

is the proportion of data points in R that have label equal to k. On the other hand, the misclassification error rate of R is defined as

$$L(R) = 1 - \max_{k \in C} \ \widehat{p}_k.$$

Recall that we use the majority label in R as the predicted label.

(b1) (1 point) Show that for binary classification with $C = \{0, 1\}$, the above metrics for any given region R reduce to

$$G(R) = 2\widehat{p}_0\widehat{p}_1, \qquad L(R) = \min\{\widehat{p}_0, \widehat{p}_1\}.$$

(b2) (2 points) Now for any given partition (R_1, R_2) due to a split of X_3, we evaluate it by using the *overall Gini index*, defined as

$$G(R_1) + G(R_2).$$

With the following labeled data points:

- Single : 20 yes, 10 no
- Married : 30 yes, 10 no
- Divorced : 5 yes, 15 no

indiacte the split of X_3 that yields the smallest overall Gini index.

(b3) (2 points) If we evaluate any given partition (R_1, R_2) by using the *overall misclassification error rate*, defined as

$$L(R_1) + L(R_2),$$

with the same labeled data points as in the previous part, indiacte the split of X_3 that yields the smallest overall misclassification error rate.

Problem 3 (8 points)

Ensemble methods combine multiple single predictors to improve the overall predictive performance. Consider a dataset where we train multiple prediction models $\{\hat{f}_1, \ldots, \hat{f}_m\}$ using m bootstrap samples. We consider the ensemble method based on bagging. The final prediction at any given x is made by averaging their outputs (for regression), that is,

$$\hat{f}_{\text{ensemble}}(x) = \frac{1}{m} \sum_{i=1}^{m} \hat{f}_i(x).$$

Assume each of the individual prediction $\hat{f}_i(x)$ satisfies:

- The variance of each model's prediction is

$$\text{Var}[\hat{f}_i(x)] = \sigma^2, \qquad \forall\ 1 \leq i \leq m.$$

- The squared bias of each prediction model is,

$$\Big(\mathbb{E}[\hat{f}_i(x)] - f^*(x)\Big)^2 = B, \qquad \forall\ 1 \leq i \leq m.$$

  Here f^* is the true regression function.

Answer the following questions.

(a) Assume the prediction models $\hat{f}_1, \ldots, \hat{f}_m$ are independent.

(a1) (1 point) Prove that

$$\mathrm{Var}[\hat{f}_{\text{ensemble}}(x)] = \frac{\sigma^2}{m}.$$

(a2) (1 point) Prove that

$$\left(f^*(x) - \mathbb{E}[\hat{f}_{\text{ensemble}}(x)]\right)^2 \leq B.$$

(a3) (3 points) Compare the expected Mean Squared Error (MSE) of $\hat{f}_{\text{ensemble}}$,

$$\mathbb{E}\left[(\hat{f}_{\text{ensemble}}(x) - f^*(x))^2\right],$$

with that of a single predictor $\hat{f}_i(x)$ and discuss the effect of m.

(b) Suppose the prediction models $\hat{f}_1, \ldots, \hat{f}_m$ are not independent but have a correlation $\rho \in [0, 1]$, that is,

$$\mathrm{Corr}(\hat{f}_i(x), \hat{f}_j(x)) = \rho, \quad \text{for all } i \neq j.$$

(b1) (1 point) Prove that in this case

$$\mathrm{Var}[\hat{f}_{\text{ensemble}}(x)] = \frac{\sigma^2}{m} + \rho\sigma^2 \left(1 - \frac{1}{m}\right).$$

(b2) (2 points) Compare the expected MSE of the ensemble prediction with that of the individual prediction $\hat{f}_i$.

Explain why reducing correlation among individual prediction models (e.g., using randomly selected feature subsets in random forests) is crucial for the success of ensemble methods.

Problem 4 (10 points, 1 point for each subquestion)

Be sure to mark your answers on the **SCANT sheet** of multiple choice questions. There is **ONLY ONE** correct answer to each question.

1. Which of the following statements is true about k-nearest neighbors (KNN)?

   A k-NN is a parametric method.

   B Increasing k reduces the variance of the k-NN prediction.

   C Cross-validation cannot be used to select the value of k.

   D k-NN assigns more weights for prediction to the closer neighbors.

2. Which of the following statements is true about support vector machines (SVM)?

   A SVM can give estimates of the posterior probabilities.

   B SVM always yields zero training error for any linearly separable data.

   C SVM aims to minimize the 0-1 misclassification error rate.

   D SVM can not be used when the data is not linearly separable.

3. Which of the following statements is true about classification decision trees?

   A Decision trees are too simple to overfit.

   B Decision trees cannot handle categorical features.

   C Increasing the depth of a decision tree always improves test accuracy.

   D Pruning a decision tree cannot decrease the training misclassification error rate.

4. Which of the following statements is true about ensemble methods?

   A Bagging always reduces both bias and variance.

   B Boosting is prone to overfitting with too many iterations.

   C Decision trees aggregated by random forests are more similar than those aggregated by bagging.

   D Ensemble methods do not work well when the feature dimension is high.

5. Which of the following statements is true about logistic regression?

   A Logistic regression is only suitable for binary classification.

   B Logistic regression can not handle categorical features.

   C Logistic regression makes less distributional assumption than discriminant analysis.

   D Regularization can not be used in the estimation of logistic regression model.

6. Which of the following statements is true about cross-validation (CV)?

   A CV can not estimate the test error of a given prediction method.

   B CV splits the dataset into non-overlapping subsets only once.

   C CV requires to specify the likelihood of the data.

   D Leave-one-out CV can only be used for linear prediction methods.

7. Which of the following statements is true about Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA)?

   A LDA assumes stronger conditions than QDA.

   B The decision boundary of QDA is always linear in the feature space.

   C LDA cannot be used when the response Y has more than two classes.

   D QDA usually has both lower bias and lower variance than LDA.

8. Which of the following statements is true about the bootstrap method?

   A The bootstrap method assumes to know the likelihood of the data.

   B Bootstrap samples are created by sampling without replacement from the original dataset.

   C The bootstrap method can be used to estimate the distribution of a statistic.

   D The number of bootstrap samples required depends directly on the sample size of the original dataset.

9. Which of the following statements is true about decision tree fitting?

   A Recursive binary splitting is a greedy approach.

   B The splitting criterion for fitting classification trees should be based on minimizing the misclassification error rate.

   C Pruning a fitted classification tree uses Gini index to select the best subtree.

   D Recursive binary splitting starts with the entire feature space and iteratively splits it into smaller non-rectangle regions.

10. Which of the following statements is true about generative and discriminative models?

A Logistic regression is a discriminative model because it models the conditional probability $\mathbb{P}(Y \mid X)$.

B Discriminant analysis is a discriminative model because it models conditional density $\mathbb{P}(X \mid Y)$.

C Discriminative models typically make more assumptions than generative models.

D Generative models can not be used for classification.